Product Requirements Document: Insurance Management SaaS Platform

Executive Summary

This document outlines the requirements for a comprehensive multi-tenant SaaS platform designed for insurance companies in Côte d'Ivoire and French-speaking regions. The platform leverages AI technology to automate policy management, streamline operations, enhance customer experience, and enable inter-company collaboration.

2.21 Point of Sale (POS) Management

2.21.1 POS Network Administration

- **Location Management**
  - Register multiple POS locations per insurance company
  - Geographic mapping of all POS locations
  - City and region-based organization
  - POS operating hours configuration
  - Contact information per location
  - Capacity and service offerings per POS
- **POS Setup & Configuration**
  - Unique POS identification codes
  - Assign managers and staff to each location
  - Product/policy type availability per POS
  - Pricing and discount rules per location
  - Territory and catchment area definition
  - Equipment and resource tracking
- **POS Performance Monitoring**
  - Real-time transaction monitoring
  - Queue management and wait time tracking
  - Customer satisfaction per location
  - Service level compliance
  - Operational efficiency metrics

2.21.2 Sales Channel Management

- **Multi-Channel Sales Tracking**
  - Point of Sale (physical locations)
  - Online Sales (client self-service portal)
  - Agent/Broker sales
  - Call center sales
  - Mobile app sales
  - Partner channel sales
- **Channel Attribution**
  - First-touch attribution
  - Last-touch attribution
  - Multi-touch attribution models
  - Assisted conversion tracking
  - Channel performance comparison

2.21.3 Sales Analytics & Reporting

POS-Level Reports:

- Daily sales summary per POS
- Weekly sales performance by location
- Monthly revenue breakdown per POS
- Yearly comparative analysis across locations
- Sales volume trends and seasonality
- Product mix analysis per POS
- Average policy value per location
- Conversion rates (quotes to policies)
- Walk-in traffic vs. sales conversion
- Peak hours and staffing optimization
- Top-performing vs. underperforming locations
- Geographic market penetration analysis

Employee Performance Reports:

- Individual sales statistics per employee
    - Daily sales count and value
    - Weekly sales achievements
    - Monthly sales targets vs. actuals
    - Yearly performance summaries
- Sales leaderboards (daily, weekly, monthly, yearly)
- Product knowledge and expertise tracking
- Quote-to-policy conversion rate per employee
- Average deal size per employee
- Customer satisfaction scores per employee
- Sales velocity (time to close)
- Cross-sell and upsell success rates
- Commission earned per employee
- Training and certification status

Online Sales Analytics:

- Self-service portal conversions
- Online quote requests
- Digital policy purchases
- Customer journey analytics
- Abandonment rate analysis
- Device and browser analytics
- Traffic source attribution
- Landing page performance
- A/B test results
- Online vs. offline sales comparison

Comparative Analytics:

- POS vs. Online sales performance
- Channel-wise revenue contribution
- Cost per acquisition by channel
- Customer lifetime value by acquisition channel
- Retention rates by sales channel
- Geographic sales heat maps
- Product popularity by region
- Demographic analysis per channel

2.21.4 Commission & Incentive Management

- Tiered commission structures
- Performance-based bonuses
- Target vs. achievement tracking
- Commission calculation automation
- Incentive program management
- Sales contests and competitions
- Reward point systems
- Commission dispute resolution
- Real-time commission visibility for employees

2.21.5 Inventory & Resource Management (POS)

- Marketing materials inventory
- Brochure and form stock management
- IT equipment tracking
- Cash management (for cash payments)
- Receipt printer supplies
- Low stock alerts
- Reorder automation
- Asset depreciation tracking

2.22 Payroll Management System

2.22.1 Employee Master Data

- **Employee Records**
  - Personal information (name, ID, contact, emergency contacts)
  - Employment details (date of hire, position, department, POS location)
  - Salary structure (base salary, allowances, benefits)
  - Bank account information for direct deposit
  - Tax identification numbers
  - Social security/pension details
  - Contract type (permanent, temporary, contract)
  - Employment history and promotions
- **Organizational Structure**
  - Department hierarchies
  - Reporting relationships
  - Cost center assignments
  - Job descriptions and salary grades
  - Position budgeting

2.22.2 Compensation Management

- **Salary Components**
  - Base salary configuration
  - Housing allowance
  - Transport allowance
  - Communication allowance
  - Meal allowance
  - Performance bonuses
  - Sales commissions (integrated with sales data)
  - Overtime pay calculation
  - Night shift differentials
  - Hazard pay (for field adjusters)
- **Deductions**
  - Income tax (progressive rates per Côte d'Ivoire regulations)
  - Social security contributions (CNPS - Caisse Nationale de Prévoyance Sociale)
  - Pension contributions
  - Health insurance premiums
  - Loan repayments
  - Advance salary recovery
  - Garnishments
  - Union dues
  - Voluntary savings schemes

2.22.3 Payroll Processing

- **Automated Payroll Cycles**
  - Monthly, bi-weekly, or weekly payroll runs
  - Automated salary calculation based on:
    - Time and attendance data
    - Sales performance (commissions)
    - Overtime hours
    - Approved leave and absences
    - Bonuses and incentives
  - Payroll preview and approval workflow
  - Payroll adjustments and corrections
  - Retroactive pay calculations
- **Payroll Execution**
  - Automated bank file generation
  - Integration with banking systems for direct deposit
  - Mobile money payout support
  - Cash payment tracking (for locations without banking)
  - Payment confirmation and reconciliation
  - Failed payment handling and retry

2.22.4 Payslip Generation & Distribution

- **Automated Payslip Features**
  - Professional payslip template (customizable)
  - Multi-language support (French primary)
  - Detailed earnings breakdown
  - Itemized deductions
  - Net pay calculation
  - Year-to-date totals
  - Leave balance summary
  - Company branding
- **Distribution Methods**
  - Email delivery (PDF format)
  - Employee self-service portal download
  - Mobile app access
  - SMS notification of payslip availability
  - Printed copies for employees without digital access
  - Archive of historical payslips
- **Payslip Security**
  - Password-protected PDF files
  - Encrypted email transmission
  - Access logs for payslip downloads
  - Tamper-evident digital signatures

2.22.5 Time & Attendance Integration

- **Time Tracking**
  - Biometric clock-in/clock-out
  - Mobile GPS-based attendance
  - Manual time entry and approval
  - Break time tracking
  - Overtime tracking and approval
  - Shift schedule management
- **Leave Management**
  - Leave type configuration (annual, sick, maternity, paternity, compassionate)
  - Leave accrual rules
  - Leave request and approval workflow
  - Leave balance tracking
  - Leave encashment calculations
  - Public holiday calendar management
  - Integration with payroll for leave deductions

2.22.6 Tax & Compliance Management

- **Tax Calculations**
  - Automated income tax computation (Côte d'Ivoire tax brackets)
  - Tax relief and allowances application
  - Monthly tax deduction and remittance
  - Annual tax certificate generation (for employees)
  - Tax reconciliation reports
- **Statutory Compliance**
  - Social security (CNPS) contribution calculation
  - Pension fund contributions
  - Regulatory reporting to government agencies
  - Year-end tax filing support
  - Compliance audit trails
- **Payroll Reports for Authorities**
  - Monthly CNPS declaration
  - Annual tax summaries
  - Labor statistics reports
  - Payroll register for audits

2.22.7 Payroll Reporting & Analytics

- **Standard Payroll Reports**
  - Payroll summary by department
  - Payroll summary by POS location
  - Cost center analysis
  - Salary distribution analysis
  - Deduction summaries
  - Bank transfer reports
  - Pay register
  - Payroll variance reports (budget vs. actual)
- **Employee Cost Analysis**
  - Total compensation cost per employee
  - Cost per POS location
  - Cost as percentage of revenue
  - Overtime cost analysis
  - Commission payout tracking
  - Benefits cost analysis
- **Compliance Reports**
  - Tax liability reports
  - Social security contribution reports
  - Pension fund reports
  - Audit reports
  - Year-end employee earnings statements

2.22.8 Employee Self-Service

- **Payroll Portal**
    - View current and historical payslips
    - Download tax certificates
    - Update bank account information
    - View leave balances
    - Submit leave requests
    - Track loan balances
    - View attendance records
    - Request salary advances
    - Update personal information
    - Beneficiary management for benefits

2.22.9 Payroll Accounting Integration

- **Automated Journal Entries**
    - Salary expense posting to general ledger
    - Deduction liability accounts
    - Net pay clearing accounts
    - Commission expense allocation
    - Department-wise cost allocation
    - Cost center distribution
- **Payroll Reconciliation**
    - Bank reconciliation for salary payments
    - Deduction payment tracking to authorities
    - Payroll accrual management
    - Audit trail for all payroll transactions

2.22.10 Advanced Payroll Features

- **Loan Management**
  - Employee loan issuance
  - Loan repayment schedules
  - Automatic deduction from salary
  - Interest calculation
  - Loan balance tracking
  - Early settlement options
- **Advance Salary**
  - Advance request and approval workflow
  - Recovery schedule setup
  - Integration with payroll deductions
- **Bonus & Incentive Programs**
  - Performance bonus calculations
  - Sales target achievement bonuses
  - Annual bonuses (13th month)
  - Profit-sharing programs
  - Retention bonuses
- **Benefits Administration**
  - Health insurance enrollment
  - Life insurance management
  - Retirement plan contributions
  - Education assistance programs
  - Transportation benefits
  - Housing loan schemes

2.22.11 Payroll Security & Audit

- **Access Controls**
  - Role-based access for payroll data
  - Sensitive data encryption
  - Payroll approval workflows
  - Change logs for all modifications
- **Audit Trails**
  - Complete history of payroll runs
  - Modification tracking
  - Approval history
  - Payment confirmation logs
  - User activity logs
- **Data Privacy**
  - GDPR compliance for employee data
  - Data retention policies
  - Right to be forgotten implementation
  - Consent management for data processing

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1. Product Overview

1.1 Vision

To create an intelligent, unified insurance management platform that empowers insurance companies to efficiently manage policies, automate workflows, provide exceptional customer service through AI assistance, and facilitate secure inter-company collaboration.

1.2 Target Users

- Insurance company administrators
- Insurance agents and employees
- Individual and corporate clients
- Multiple insurance companies (multi-tenant architecture)

1.3 Market Context

- **Primary Market**: Côte d'Ivoire
- **Default Language**: French
- **Secondary Languages**: English, local languages (expandable)
- **Insurance Types**: Vehicle, Property/House, Life, Health, Business, Travel, Agriculture

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2. Core Features & Requirements

2.1 User Management & Authentication

2.1.1 Role-Based Access Control (RBAC)

- **Super Admin**: Platform-level management across all insurance companies
- **Company Admin**: Full control within their insurance company
- **Manager**: Department-level oversight and approvals
- **Agent/Staff**: Client-facing operations, policy creation
- **Client**: Self-service portal access

2.1.2 Authentication & Security

- Multi-factor authentication (MFA)
- Single Sign-On (SSO) capability
- Biometric authentication for mobile apps
- Session management with auto-logout
- Password policies and rotation requirements
- OAuth integration for social login
- Audit logs for all user activities

2.1.3 Permission Management

- Granular permissions at feature and data level
- Custom role creation capability
- Permission inheritance and override
- Department-based access restrictions
- Field-level data access control
- Approval workflows for sensitive operations

2.2 Client Management

2.2.1 Client Portal Features

- Self-service account creation and management
- Profile management (personal, contact, beneficiary information)
- Document upload center with drag-and-drop
- Policy overview dashboard
- Claims history and status tracking
- Payment history and upcoming premiums
- Renewal reminders and management
- Referral program dashboard

2.2.2 Client Database Management

- Comprehensive client profiles (individuals and businesses)
- Contact information management
- Family/dependent linking
- Risk assessment profiles
- Communication preference settings
- Document repository per client
- Interaction history tracking
- 360-degree client view
- Client segmentation for marketing
- Duplicate detection and merging
- GDPR/data privacy compliance tools

2.2.3 KYC (Know Your Customer) Management

- Digital identity verification
- Document verification workflows
- Risk scoring and categorization
- Compliance checklist automation
- Sanctions and watchlist screening
- Periodic KYC renewal triggers

2.3 Policy/Contract Management

2.3.1 Policy Types Supported

- Vehicle Insurance (Auto, Motorcycle, Fleet)
- Property Insurance (Home, Building, Rental)
- Life Insurance (Term, Whole Life, Universal)
- Health Insurance (Individual, Family, Group)
- Business Insurance (Liability, Property, Professional)
- Travel Insurance
- Agricultural Insurance

2.3.2 Quote Generation System

- AI-powered quote calculator based on:
    - Client information and risk profile
    - Asset details and valuation
    - Coverage options and limits
    - Historical claims data
    - Market pricing intelligence
- Real-time quote generation
- Multiple quote comparison view
- Quote validity period management
- Quote revision and versioning
- Instant quote delivery (email, SMS, WhatsApp)
- Quote-to-policy conversion workflow

2.3.3 Policy Lifecycle Management

- Automated policy creation from accepted quotes
- Policy document generation with QR codes
- Policy numbering and versioning
- Policy binding and issuance
- Mid-term adjustments and endorsements
- Renewal management with automation
- Cancellation and lapse processing
- Policy reinstatement workflows
- Policy transfer between clients
- Beneficiary management

2.3.4 Automated Policy Assignment

- AI-driven policy matching based on:
  - Client demographics and needs
  - Uploaded documents (ID, vehicle registration, property deeds)
  - Risk assessment scores
  - Budget and coverage preferences
  - Historical purchasing behavior
- Intelligent product recommendation engine
- Cross-sell and upsell suggestions
- Bundle creation and discounting

2.3.5 Contract Template Management

- Customizable policy templates per insurance type
- Dynamic field insertion
- Multi-language template support
- Version control for templates
- Legal clause library
- Terms and conditions management

2.4 Document Management System

2.4.1 Document Processing

- Automated document classification using AI/OCR
- Intelligent data extraction from documents:
  - ID cards and passports
  - Driver's licenses
  - Vehicle registration documents
  - Property titles and deeds
  - Medical records
  - Income statements
  - Business registration documents
- Document validation and verification
- Quality checks (resolution, completeness)
- Automatic filing in appropriate client folders
- Version control and audit trails

2.4.2 Document Storage & Retrieval

- Secure cloud-based storage
- Hierarchical folder structure
- Advanced search capabilities (full-text, metadata, AI-semantic search)
- Document tagging and categorization
- Quick access to frequently used documents
- Archival and retention policy automation
- Document expiry tracking and alerts

2.4.3 Inter-Company Document Sharing

- Granular sharing controls per document
- Share with specific insurance companies
- Time-limited access grants
- Watermarking for shared documents
- Access logs and download tracking
- Revocation capabilities
- Secure document transfer protocols
- Request system for document access

2.4.4 Electronic Signatures

- Digital signature integration (e-signature platforms)
- Signature workflow management
- Multi-party signing support
- Certificate-based signatures
- Biometric signature capture
- Signature audit trails
- Legal compliance (eIDAS, ESIGN Act)

2.5 AI Assistant Integration

2.5.1 Voice-Enabled AI Assistant

- Multi-language voice recognition (French primary)
- Natural language understanding
- Voice commands for:
  - Policy inquiries
  - Claims status checks
  - Document retrieval
  - Payment information
  - Appointment scheduling
- Voice-to-text transcription for records
- Integration with mobile and web applications

2.5.2 AI-Powered Customer Guidance

- Intelligent chatbot for client self-service
- Policy recommendation based on needs assessment
- Coverage gap analysis
- Premium calculation assistance
- Claims guidance and documentation help
- FAQ automation with learning capabilities
- Sentiment analysis for customer satisfaction
- Escalation to human agents when needed

2.5.3 AI for Staff & Agents

- Information retrieval assistant
- Policy comparison and analysis
- Client history summarization
- Risk assessment assistance
- Document verification support
- Underwriting decision support
- Next-best-action recommendations
- Sales coaching and script assistance

2.5.4 Advanced AI Capabilities

- Predictive analytics for:
  - Claims likelihood prediction
  - Customer churn prediction
  - Premium optimization
  - Fraud detection
- Automated underwriting for standard cases
- Natural language query of database
- Intelligent document summarization
- Anomaly detection in claims and applications
- Computer vision for damage assessment (photos/videos)

2.6 Claims Management

2.6.1 Claims Submission & Tracking

- Multi-channel claim submission (portal, mobile, phone, email)
- Guided claims form with AI assistance
- Photo/video evidence upload
- Real-time claims status tracking
- Automated acknowledgment and reference number generation
- Estimated settlement timeline

2.6.2 Claims Processing Workflow

- Automated claims routing and assignment
- Adjustor assignment and scheduling
- Document verification checklists
- Investigation case management
- Approval workflow with multi-level authorization
- Payment processing integration
- Claims diary and activity logging

2.6.3 Fraud Detection

- AI-powered fraud scoring
- Pattern recognition across claims
- Anomaly detection algorithms
- Duplicate claims detection
- Cross-referencing with external databases
- Investigation flagging and workflows
- Reporting and analytics on fraud trends

2.7 Financial Management

2.7.1 Premium Management

- Premium calculation engine
- Payment schedule generation
- Grace period management
- Late payment tracking and penalties
- Premium refund processing
- Commission calculation and tracking

2.7.2 Multi-Payment Gateway Integration

- **International**: Stripe, PayPal
- **Local Mobile Money**: Orange Money, MTN Mobile Money, Moov Money, Wave
- **Bank Transfers**: Direct bank integration
- Payment method management per client
- Recurring payment setup (auto-debit)
- Payment confirmation and reconciliation
- Failed payment retry logic
- Refund processing

2.7.3 Client Payment Management

- **Monthly Payment Tracking**
  - Automated payment schedule tracking per client
  - Payment due date management
  - Payment history and audit trail
  - Partial payment handling
  - Payment allocation across multiple policies
  - Outstanding balance tracking
- **Missed Payment Management**
  - Automated reminder system for upcoming payments (7 days, 3 days, 1 day before)
  - Missed payment notifications (email, SMS, WhatsApp, push)
  - Escalating reminder sequences (1 day, 3 days, 7 days, 14 days overdue)
  - Grace period tracking and enforcement
  - Late payment fee calculation and application
  - Payment plan restructuring for overdue accounts
  - Suspension and reinstatement workflows
  - Collections process automation
- **Payment Convenience Features**
  - One-click payment from reminders
  - Multiple payment method options
  - Payment link generation
  - QR code payment options
  - Recurring payment enrollment
  - Payment receipt generation and delivery

2.7.4 Financial Accounting System

- **Chart of Accounts**
  - Customizable account structure
  - Multi-currency support (XOF primary)
  - Account hierarchies and groupings
  - Cost center and department allocation
- **General Ledger**
  - Automated journal entry creation
  - Double-entry bookkeeping
  - Transaction posting and reconciliation
  - Period-end closing procedures
  - Audit trail for all transactions
- **Accounts Receivable**
  - Premium receivables tracking
  - Aging reports (30, 60, 90, 120+ days)
  - Customer credit management
  - Collections tracking
  - Write-off management
- **Accounts Payable**
  - Claims payable tracking
  - Commission payable management
  - Vendor management
  - Payment processing workflows
  - Expense tracking
- **Bank Reconciliation**
  - Automated bank statement import
  - Transaction matching algorithms
  - Discrepancy identification
  - Reconciliation reports
  - Multi-bank account support
- **Fixed Assets Management**
  - Asset registration and tracking
  - Depreciation calculation (straight-line, declining balance)
  - Asset disposal and write-offs

- Asset valuation reports

2.7.5 Financial Reporting

- **Revenue Reports**
  - Daily revenue summary
  - Weekly revenue trends
  - Monthly revenue statements
  - Quarterly financial reports
  - Annual financial statements
  - Revenue by product line
  - Revenue by geographic location
  - Revenue by sales channel (online vs. point of sale)
  - Revenue forecasting and projections
- **Operational Reports**
  - Premium collection reports
  - Claims payout analytics
  - Loss ratio calculations
  - Combined ratio analysis
  - Commission statements
  - Accounts receivable aging
  - Cash flow statements
  - Profit and loss statements
  - Balance sheet
  - Trial balance
- **Compliance Reports**
  - Regulatory financial submissions
  - Solvency margin reports
  - Statutory accounting reports
  - Tax reports (VAT, corporate tax)
  - Audit preparation reports
- **Financial Dashboards**
  - Real-time financial KPIs
  - Revenue vs. target tracking
  - Collection efficiency metrics
  - Outstanding premium visualization
  - Cash position monitoring
  - Integration with accounting software (QuickBooks, Sage, OHADA standards)

- Integration with accounting software (QuickBooks, Sage, OHADA standards)

2.8 Task & Workflow Management

2.8.1 Employee Task Management

- Task assignment and tracking
- Priority and deadline management
- Task dependencies and sequencing
- Automated task creation from triggers
- Workload balancing
- Task templates for common workflows
- Reminders and notifications
- Performance tracking

2.8.2 Approval Workflows

- Configurable multi-step approval processes
- Role-based approval routing
- Parallel and sequential approval support
- Approval history and audit trails
- Delegation and substitution capabilities
- Timeout and escalation rules
- Mobile approval capabilities

2.8.3 Process Automation

- Business rules engine
- Scheduled jobs and batch processing
- Event-driven automation triggers
- Integration with external systems via webhooks
- Workflow orchestration
- SLA monitoring and enforcement

2.9 Communication Systems

2.9.1 Internal Chat System

- Real-time messaging between staff
- Group chats and channels
- Direct messaging
- File sharing in conversations
- Message search and archival
- Presence indicators (online/offline)
- Read receipts
- Mobile and desktop notifications

2.9.2 Client Communication

- Secure messaging with clients
- Automated notifications (email, SMS, push, WhatsApp)
- Communication templates
- Scheduled communications
- Communication history per client
- Bulk messaging capabilities
- Opt-in/opt-out management
- Multi-channel orchestration

2.9.3 Inter-Company Communication

- Secure messaging between insurance companies
- Co-insurance and reinsurance coordination
- Information request workflows
- Shared case discussions
- Conference call integration
- Meeting scheduling

2.10 Ticket/Support System

2.10.1 Ticketing Features

- Multi-channel ticket creation (email, portal, chat, phone)
- Automated ticket categorization and routing
- Priority and severity levels
- SLA tracking and breach alerts
- Internal notes and collaboration
- Customer satisfaction surveys post-resolution
- Knowledge base integration

2.10.2 Complaint Management

- Dedicated complaint tracking
- Regulatory reporting requirements
- Escalation procedures
- Resolution tracking and timelines
- Root cause analysis
- Trending and pattern identification

2.10.3 Suggestion Management

- Idea submission system
- Voting and prioritization
- Status tracking (under review, planned, implemented)
- Feedback loop to submitters

2.11 Reporting & Analytics

2.11.1 Standard Reports

- Policy portfolio analysis
- Sales performance reports
- Claims analysis and trends
- Customer acquisition and retention
- Agent productivity reports
- Financial performance dashboards
- Regulatory compliance reports
- Risk exposure analysis

2.11.2 Custom Report Builder

- Drag-and-drop report designer
- Custom fields and calculated metrics
- Multiple visualization types (charts, tables, maps)
- Scheduled report generation and distribution
- Export to multiple formats (PDF, Excel, CSV)
- Report sharing and collaboration

2.11.3 Business Intelligence

- Interactive dashboards
- Drill-down capabilities
- Predictive analytics
- Benchmarking against industry standards
- Real-time data visualization
- Mobile BI access

2.12 Referral & Loyalty Program

2.12.1 Referral System

- Client referral tracking
- Unique referral codes/links
- Referral reward management
- Multi-tier referral bonuses
- Social sharing integration
- Referral analytics and conversion tracking

2.12.2 Loyalty & Discount Management

- Points-based loyalty program
- Tiered customer benefits
- Promotional discount campaigns
- Bundle discounts
- Renewal discounts for claim-free periods
- Volume-based discounts
- Special occasion offers (birthdays, anniversaries)
- Coupon code management

2.13 Multi-Tenant Architecture

2.13.1 Company Management

- Self-service company onboarding
- Company profile and branding customization
- Subscription and billing management
- Feature toggle per company
- Data isolation and security
- Company-specific configurations
- White-label capability

2.13.2 Inter-Company Features

- Company directory and network
- Collaboration requests and approvals
- Co-insurance management
- Reinsurance tracking
- Information exchange protocols
- Shared blacklist/watchlist
- Industry-wide analytics (anonymized)

2.14 Integration & API Management

2.14.1 API Platform

- RESTful API for all core functions
- GraphQL endpoint for flexible queries
- Webhook support for event-driven integrations
- API documentation (Swagger/OpenAPI)
- API versioning and deprecation management
- Rate limiting and throttling
- API key and OAuth management
- Sandbox environment for testing

2.14.2 Third-Party Integrations

- **CRM Systems**: Salesforce, HubSpot
- **Accounting Software**: QuickBooks, Xero, Sage
- **Communication Platforms**: Twilio (SMS), SendGrid (Email), WhatsApp Business
- **Identity Verification**: Onfido, Jumio
- **Credit Bureaus**: Local and international
- **Government Databases**: Vehicle registration, land registry
- **Telematics**: GPS and driving behavior data
- **Weather Data**: For risk assessment
- **Medical Networks**: For health insurance claims
- **Repair Networks**: Garages, contractors

2.15 Mobile Application

2.15.1 Mobile Features

- Native iOS and Android applications
- All core portal features optimized for mobile
- Offline capability for essential functions
- Push notifications
- Biometric authentication
- Camera integration for document capture
- GPS for location-based services
- Voice assistant integration
- Digital insurance card/wallet

2.16 Notification System

2.16.1 Notification Types

- Policy renewals and expirations
- Payment reminders and receipts
- Claims status updates
- Document requirements
- Task assignments and deadlines
- System announcements
- Promotional offers
- Compliance deadlines

2.16.2 Notification Channels

- Email
- SMS
- Push notifications (mobile and web)
- WhatsApp Business
- In-app notifications
- Phone calls (automated voice)

2.16.3 Notification Management

- User preference center
- Frequency controls (immediate, daily digest, weekly)
- Channel preferences per notification type
- Do-not-disturb hours
- Notification templates
- A/B testing for communication effectiveness

2.17 Compliance & Regulatory

2.17.1 Data Protection

- GDPR compliance tools
- Data retention policies
- Right to be forgotten implementation
- Data portability features
- Consent management
- Privacy policy acceptance tracking
- Data breach notification procedures

2.17.2 Insurance Regulations

- Côte d'Ivoire insurance regulations compliance
- CIMA (Conference Interafricaine des Marchés d'Assurances) requirements
- Solvency reporting
- Regulatory filing automation
- Audit trail for all transactions
- Policy form approval tracking

2.17.3 Security & Compliance

- ISO 27001 compliance
- SOC 2 Type II certification
- Encryption at rest and in transit
- Regular penetration testing
- Vulnerability management
- Disaster recovery and business continuity
- Backup and restore procedures

2.18 Social Media Integration

2.18.1 Social Sharing Features

- Share insurance products on social platforms
- Referral link sharing
- Educational content distribution
- Success stories and testimonials
- Quote request from social media
- Social login integration

2.18.2 Social Media Management

- Post scheduling
- Engagement tracking
- Social listening for brand mentions
- Lead generation from social channels
- Chatbot integration (Facebook Messenger, etc.)

2.19 QR Code & Document Verification

2.19.1 QR Code Generation

- Unique QR codes on all policy documents
- Encode policy details and verification URL
- Digital certificate integration
- Tamper-evident features

2.19.2 Verification System

- Public verification portal
- Mobile app QR scanner
- Instant policy authenticity check
- Access to basic policy information
- Verification audit logs
- Fraud prevention

2.20 Advanced Features

2.20.1 Telematics Integration

- Usage-based insurance (UBI) support
- Real-time driving behavior monitoring
- Mileage tracking
- Accident detection and emergency response
- Safe driving rewards

2.20.2 IoT Device Integration

- Smart home device data for property insurance
- Health wearables for life/health insurance
- Connected vehicle data
- Risk mitigation through real-time monitoring

2.20.3 Blockchain Features

- Immutable policy records
- Smart contracts for automated claims
- Transparent claims processing
- Fraud prevention through distributed ledger

2.20.4 Parametric Insurance

- Weather-based automatic payouts
- Index-based triggers
- No claims process required
- IoT sensor integration

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3. Technical Requirements

3.1 Architecture

- Microservices architecture for scalability
- Cloud-native design (AWS, Azure, or Google Cloud)
- Multi-tenant database architecture with data isolation
- Containerization (Docker, Kubernetes)
- Serverless functions for specific workloads
- Event-driven architecture
- CDN for global content delivery

3.2 Performance Requirements

- Page load time < 2 seconds
- API response time < 500ms for 95th percentile
- Support 100,000+ concurrent users
- 99.9% uptime SLA
- Real-time data synchronization
- Horizontal scalability

3.3 Technology Stack Recommendations

- **Backend**: Node.js, Python (Django/FastAPI), or Java (Spring Boot)
- **Frontend**: React.js or Vue.js
- **Mobile**: React Native or Flutter
- **Database**: PostgreSQL (primary), MongoDB (documents), Redis (cache)
- **Search**: Elasticsearch
- **Message Queue**: RabbitMQ or Apache Kafka
- **AI/ML**: TensorFlow, PyTorch, OpenAI API, Anthropic Claude API
- **OCR**: Tesseract, Google Vision API, AWS Textract
- **Storage**: AWS S3, Azure Blob Storage, or Google Cloud Storage

3.4 Security Requirements

- End-to-end encryption
- WAF (Web Application Firewall)
- DDoS protection
- Regular security audits
- Penetration testing (quarterly)
- Intrusion detection system
- Security information and event management (SIEM)

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4. User Experience Requirements

4.1 Design Principles

- Mobile-first responsive design
- Accessibility (WCAG 2.1 AA compliance)
- Intuitive navigation and information architecture
- Consistent branding across all touchpoints
- Progressive disclosure of information
- Error prevention and helpful error messages
- Loading indicators for all async operations

4.2 Localization

- French as default language
- Support for English and local languages
- Right-to-left (RTL) support for future expansion
- Currency and date format localization
- Cultural considerations in UI/UX
- Local payment method prominence

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5. Implementation Phases

Phase 1: Foundation (Months 1-4)

- Core user management and authentication
- Basic client management
- Policy creation and management
- Document upload and storage
- Payment gateway integration
- Admin dashboard
- Basic POS location setup
- Employee master data management

Phase 2: Intelligence (Months 5-7)

- AI assistant integration (basic)
- Automated document processing
- Quote generation engine
- Basic reporting and analytics
- Mobile application (MVP)
- POS sales tracking
- Financial accounting foundation

Phase 3: Collaboration & Operations (Months 8-10)

- Chat and communication systems
- Inter-company features
- Advanced AI capabilities
- Claims management
- Ticket system
- Enhanced reporting
- Payroll management system
- Time and attendance integration
- Client payment tracking and reminders

Phase 4: Optimization & Analytics (Months 11-12)

- Advanced automation and workflows
- Referral and loyalty programs
- Social media integration
- Performance optimization
- Security hardening
- Beta testing and refinement
- Advanced POS analytics
- Employee performance dashboards
- Comprehensive financial reporting

Phase 5: Launch & Scale (Month 13+)

- Production launch
- User training and onboarding
- Continuous improvement
- Advanced features rollout
- Market expansion
- Payroll compliance and statutory reporting

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6. Success Metrics

6.1 Business Metrics

- Number of insurance companies onboarded
- Total policies under management
- Premium volume processed
- Customer satisfaction (NPS score)
- Policy conversion rate from quotes
- Customer retention rate
- Average time to policy issuance
- Claims processing time reduction
- Number of active POS locations
- Revenue per POS location
- Online vs. offline sales ratio
- Employee productivity index
- Payroll processing accuracy rate
- Payment collection efficiency

6.2 Technical Metrics

- System uptime percentage
- Average API response time
- Error rate
- User adoption rate per feature
- Mobile app downloads and active users
- AI assistant usage and success rate
- Payroll processing time
- Financial report generation speed

6.3 User Engagement Metrics

- Daily/Monthly active users
- Session duration
- Feature utilization rates
- Support ticket volume trends
- User feedback scores
- Employee self-service portal adoption
- Client payment portal usage

6.4 Financial Metrics

- Revenue growth rate
- Collection efficiency ratio
- Outstanding premium percentage
- Days sales outstanding (DSO)
- Payroll cost as percentage of revenue
- Commission payout accuracy
- Online payment success rate

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7. Risk Management

7.1 Technical Risks

- Scalability challenges with rapid user growth
- Integration complexity with legacy systems
- Data migration issues
- Security vulnerabilities

Mitigation: Robust architecture planning, phased rollout, comprehensive testing, security-first approach

7.2 Business Risks

- Regulatory changes
- Market competition
- User adoption resistance
- Data privacy concerns

Mitigation: Legal compliance team, competitive analysis, user training programs, transparent privacy policies

8. Support & Maintenance

8.1 Customer Support

- 24/7 support availability
- Multi-channel support (phone, email, chat)
- Tiered support levels
- Knowledge base and self-service resources
- Video tutorials and webinars
- Community forum

8.2 System Maintenance

- Regular updates and patches
- Database optimization
- Performance monitoring
- Backup verification
- Disaster recovery testing
- Feature enhancements based on feedback

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9. Budget & Resources

9.1 Development Team Requirements

- Product Manager
- UI/UX Designers (2)
- Frontend Developers (4)
- Backend Developers (6)
- Mobile Developers (3)
- AI/ML Engineers (2)
- DevOps Engineers (2)
- QA Engineers (3)
- Security Specialist
- Technical Writer
- Financial Systems Specialist
- Payroll Compliance Specialist

9.2 Budget Estimate

- Development: $900,000 - $1,400,000
- Infrastructure (Year 1): $150,000 - $250,000
- Third-party services and licenses: $120,000 - $220,000
- Marketing and sales: $200,000 - $300,000
- **Total Initial Investment**: $1,370,000 - $2,170,000

9.3 Ongoing Costs (Monthly)

- Infrastructure and hosting: $25,000 - $45,000
- Third-party APIs and services: $12,000 - $25,000
- Support and maintenance: $35,000 - $60,000
- Marketing: $15,000 - $30,000
- Compliance and legal: $5,000 - $10,000

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10. Conclusion

This Insurance Management SaaS platform represents a comprehensive solution that leverages modern AI technology to transform how insurance companies operate in Côte d'Ivoire and beyond. By automating workflows, enhancing customer experience, and enabling collaboration, the platform positions insurance companies to compete effectively in the digital age while maintaining regulatory compliance and data security.

The phased implementation approach ensures manageable development cycles while delivering value incrementally. Success will be measured through adoption rates, operational efficiency gains, customer satisfaction, and the platform's ability to scale across multiple insurance companies and markets.

Appendix A: Glossary

- **KYC**: Know Your Customer
- **OCR**: Optical Character Recognition
- **QR Code**: Quick Response Code
- **SaaS**: Software as a Service
- **SLA**: Service Level Agreement
- **API**: Application Programming Interface
- **RBAC**: Role-Based Access Control
- **NPS**: Net Promoter Score
- **CIMA**: Conférence Interafricaine des Marchés d'Assurances
- **UBI**: Usage-Based Insurance
- **IoT**: Internet of Things
- **POS**: Point of Sale
- **CNPS**: Caisse Nationale de Prévoyance Sociale (Social Security Fund)
- **DSO**: Days Sales Outstanding
- **XOF**: West African CFA Franc
- **OHADA**: Organisation pour l'Harmonisation en Afrique du Droit des Affaires

Appendix B: Compliance References

- GDPR (General Data Protection Regulation)
- CIMA Insurance Regulations
- Côte d'Ivoire Data Protection Laws
- ISO 27001 (Information Security Management)
- SOC 2 Type II (System and Organization Controls)

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